| | PHASE 1 (Fragile)<br>**Lecithotrophic Phase**<br>(0-3 dpe) | PHASE 2 (Transition)<br>**Live Feed Transition**<br>(4-12 dpe) | PHASE 3 (Adapted)<br>**Post-flexion / Weaning**<br>(15+ dpe) |
|---|---|---|---|
| **FEED TYPE** | Yolk reserves<br>(endogenous) | Rotifers + Artemia<br>(live prey introduction) | Formulated feed<br>(inert diet) |
| **MICROBIAL SUCCESSION**<br>*(Li et al., 2026)* | Uncolonized /<br>Low microbial density | Pseudoalteromonas<br>→ Stenotrophomonas<br>→ Cohaesibacter | Psychrobacter dominance<br>(core genus from day 19) |
| **MUCOSAL EFFECTORS** | UBIM / S30FAU<br>(maternal transcripts) | LEG3, CCL11, CCL20<br>(lectins + chemokines) | PA2G1B, piscidins (moro1),<br>β-defensins, HAMP2 |
| **GUT REGIONAL SPECIALIZATION**<br>*(Calduch-Giner 2016; Xue 2026)* | Undifferentiated<br>digestive tube | Midgut mucosal<br>differentiation | Anterior/Midgut: Nutrients<br>Posterior: Defense, BEST4+ &<br>lysosome-rich enterocytes |

**KEY TRANSITIONS & ONTOGENETIC MILESTONES:**

- **Mouth opening (3–4 dpe):**
First exogenous feed intake and initiation of environmental microbial exposure.

- **Live feed transition (4–12 dpe):**
Active bacterial succession and recruitment of chemokines/lectins for mucosal defense.

- **Formulated feed & weaning (15+ dpe):**
*Psychrobacter* core dominance, expression of mature piscidins (moro1), and gut compartmentalization.